ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice e.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Symbols and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 Any FIVE devices with the energy produced ✓✓✓✓✓: motor – mechanical (kinetic) energy; lamp / LED – light; buzzer / speaker – sound; heater / soldering iron – heat; solenoid – linear movement; battery charger – chemical energy. (5)

1.2 Input: what goes into the system ✓ – money and the drink selection ✓. Process: what the system does with the input ✓ – checks the money, selects the drink, calculates change ✓. Output: the result ✓ – the drink and the change are delivered ✓. (6)

1.3 A bimetal strip is made of two metals that expand at different rates ✓ bonded together ✓; when it heats up, one metal expands more and the strip bends ✓; at the set temperature it bends far enough to open the contacts and switch off the heater ✓. (4)

1.4 A gearbox transmits power from a motor to a load ✓ and changes the speed and torque ✓ (usually reducing speed and increasing torque) and can change the direction of rotation ✓. Gear for 90° ✓✓: bevel gears (or worm and wheel). (5)

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QUESTION 2

2.1.1 A device that detects a physical quantity (e.g. position, light, temperature) ✓ and changes it into a signal for the controller ✓ (2)

2.1.2 A device that receives a signal from the controller ✓ and produces movement or another action (e.g. motor, cylinder, solenoid) ✓ (2)

2.1.3 An electronic circuit with one or more inputs and one output ✓ whose output follows a logic rule (AND, OR, NOT) using two states, 0 and 1 ✓ (2)

2.2 Drawing: S1 and S2 in series with E1 across the supply ✓✓. The lamp lights only when S1 AND S2 are both closed ✓; if either switch is open, no current flows ✓. (4)

2.3.1 Correct NO push-button symbol: contact open, push-button actuator ✓✓ (2)

2.3.2 Correct NC push-button symbol: contact closed, push-button actuator ✓✓ (2)

2.3.3 Switch contact with a roller actuator ✓✓ (2)

2.4.1 Programmable Logic Controller ✓

2.4.2 Human–Machine Interface ✓

2.4.3 Personal Protective Equipment ✓

2.4.4 Direct Current ✓

(4)

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QUESTION 3

3.1 Any FOUR advantages ✓✓✓✓: air is freely available; clean (no oil leaks on products); safe in explosive areas (no sparks); fast movement; easy to store in a receiver; overload-safe; simple components. Any TWO disadvantages ✓✓: noisy exhaust; air must be prepared (filtered, dried); limited force; air is compressible, so exact speed and position control is difficult; compressing air costs energy. (6)

3.2 Filter ✓ – removes dirt and water from the air ✓; pressure regulator (with gauge) ✓ – keeps the working pressure constant at the set value ✓; lubricator ✓ – adds a fine oil mist to lubricate the components ✓ (6)

3.3 Cylinder barrel with piston and rod ✓, one air port ✓, spring drawn on the rod side ✓ (3)

3.4 It lets a cylinder or valve be operated from either of two places (OR function) ✓✓, while blocking the other input so that air does not escape through it ✓. (3)

3.5 $\mathrm{F} = \mathrm{p} \times \mathrm{A} = 600\,000 \times 0{,}002\ ✓ = 1\,200\ \mathrm{N}\ ✓$ (2)

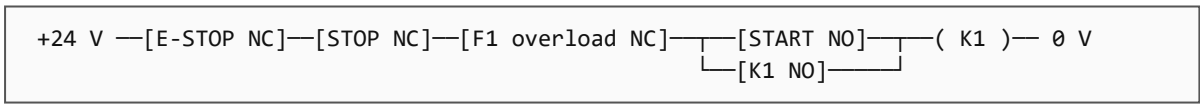
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QUESTION 4

4.1 A fuse has a thin wire that melts when the current is too high ✓ and must then be replaced ✓. A circuit breaker opens its contacts when the current is too high ✓ and can be reset and used again ✓. (4)

4.2 It switches off the supply when current leaks to earth (e.g. through a person) ✓, preventing electric shock and fires ✓. (2)

4.3 Supply lines ✓; E-stop (NC) ✓, STOP (NC) ✓ and overload contact (NC) ✓ in series; START (NO) ✓ with K1 holding contact in parallel ✓; contactor coil K1 ✓; correct symbols, neat ✓, e.g. (8)



4.4 It is a safe, extra-low voltage that greatly reduces the risk of electric shock ✓ and suits sensors, PLCs and solenoid valves ✓. (2)

4.5 Any FOUR ✓✓✓✓: switch off and isolate the supply; lock out and tag the isolator; test that the circuit is dead with a tester; wear the correct PPE; use insulated tools; make sure your hands and the floor are dry; get permission / work with a supervisor. (4)

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QUESTION 5

5.1 Any FOUR ✓✓✓✓: worn or leaking piston seals; damaged or bent piston rod; worn rod seal / wiper; air leaks at the ports or fittings; loose mountings; sticking caused by dirt or lack of lubrication; damaged cushioning. (4)

5.2 The system is divided into two halves and a test is done at the middle point ✓; this shows which half contains the fault ✓; that half is then split again ✓, and so on, until the faulty stage is found with the fewest tests ✓. (4)

5.3 Start/end – oval (rounded rectangle) ✓; process – rectangle ✓; decision – diamond ✓; input/output – parallelogram ✓ (4)

5.4 Preventive maintenance is planned and done regularly to stop breakdowns from happening ✓✓ (e.g. lubricating, draining water traps). Corrective maintenance is done after a fault or breakdown to repair the equipment ✓✓. (4)

5.5 Any FOUR ✓✓✓✓: machine name / number; date; work done; parts replaced; fault found; time taken / downtime; name and signature of the technician; date of the next service. (4)

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TOTAL: 100

END OF MARKING GUIDELINE